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T2 · 13:45-15:30

ROOM	Grand Ballroom A&B	204	205	206	207
13:45-15:30	[S25] Practical experiences and lessons from Korea's long-span sea-crossing bridges O&M	[S05] Physics-informed machine learning for structural and concrete systems	[G07-2] Advanced structural materials: Fiber-reinforced polymer and composites	[G08-4] State-of-the-art design and analysis: Probabilistic methods for optimization and data-driven assessment	[G06-6] Innovative structures and construction technologies: Analysis and design of energy, offshore, and special-purpose structures
Session Chair	Young Min Jin	Hyun Woo Park, Dongho Jeon	Roneh Glenn Libre Jr.	Rade Hajdin	Seungjun Kim
13:45	Governance and Technological Advancements in Long-Span Sea-Crossing Bridges: A Decade of Korea's Sea-Crossing Bridge Safety Division <i>Lee KeonHee</i>	Deep Learning-Enhanced Complete Quadratic Combination Method for Seismic Response Estimation <i>Dongjin Kim</i>	Simplified Material Development Procedures for Bamboo Fiber Textile <i>Roneh Glenn Libre Jr.</i>	Hierarchical Multi-Objective Optimization for Hybrid Variable Design of Composite Structures: A Case Study on Tubular Deployable Booms <i>Xiao Liang</i>	Global Hydrodynamic Response of Multi-Chord Trussed Submerged Floating Tunnels <i>Jaemin Kim</i>
14:00	Application of MTMD for Vortex-Induced Vibration of a Cable-Stayed Bridge Girder : A Maintenance Case Study <i>Han Gal Jeong</i>	Physics-Informed Neural Operators for Vibro-Acoustic Coupling in Urban Rail Transit Bridges <i>Hualun Zhou</i>	Flexural Characteristics of Ferrocement Composites: Insights from Preliminary Beam Tests <i>Mhd Rony Asshidiqie</i>	From Diagnosis to Safety Assessment: Integrated Methodology for Historic Timber Floors <i>Martins Petersons</i>	Reduction of Seismic Waves with an Optimized System of Meta-Blocks <i>Mau Nhat An Nguyen</i>
14:15	A Case Study on Maintenance of the Closed Longitudinal Corrosion of Cable Stayed Bridge with Steel Plate Deck <i>SeungHo Yeom</i>	Physics-Based Transformer Model for Data-Efficient Prediction of Nonlinear Hysteresis Across Varying Structural Properties <i>Jinseo Park</i>	Shear Capacity of Post-Tensioned Concrete Beams Considering the Influence of Prestressing Ducts <i>Dong Xu</i>	Data-Driven Multi-Constraint Optimization Design of Super-Tall Residential Braced Steel Frame Structures Based on Incremental Sensitivity Analysis <i>Yutong Xu</i>	Second-Order Wave Load Effects on a Floating Bridge Using Frequency-Domain Multimode Spectral Analysis <i>Soomin Kim</i>
14:30	Non-Entry Vertical Scanning System and AI-Based Multi-Damage Detection for Inspection of Vertical Tunnel Facilities <i>Hoon Yoo</i>	Inverse Design Optimization of High-Temperature Recycled Aggregate Concrete Using Tabular Generative Adversarial Networks and Gene Expression Programming <i>Hongrui Li</i>	Evaluating the Structural Performance of Bambusa Blumeana J-Bolt P-Connections Against Tensile Loads: Experimental Findings and Design Implications <i>Jomari Tan</i>	Economical Design of Reinforced Concrete Cantilever Retaining Wall Using Metaheuristic Algorithms <i>Mary Joanne Aniñon</i>	Ultimate Strength Evaluation of Nuclear Power Plant Shear Wall with Openings <i>Jaehan Oh</i>
14:45	Lessons from the Ulsan Bridge Fire: Toward a Risk-Based Management <i>Chan Min Park</i>	Complex-Frequency-Driven Wave-Based Analog Computing Beyond Passivity Constraints <i>Moonyeup Shin</i>	Investigation of the Residual Compressive Strength of Ferrocement-Reinforced Concrete Elements Pre-Exposed to Marine Environment and Subsequent High Temperature <i>Mary Joy Angulo</i>	Application of Deep Learning Domain Adaptation in Structural Dynamic Feature Extraction <i>Geonwoo Kim</i>	A Nonlinear Finite Element Analysis of Steel-Plate Concrete(SC) Walls for Combined-Force Based Safety Margin Evaluation in Nuclear Power Plants <i>Hyukjin Kwon</i>
15:00	Introduction to a Digital Sensor-Based Monitoring System for Cable-Supported Bridges <i>Seung Ho Yeom</i>	Physics-Informed Machine Learning for Crack Assessment in EDGE-Cracked Beams Using PZT-Based Scattering Coefficients <i>Nur Indah Mukharromah</i>		Feasibility Study on the Application of Bridge Weigh-In-Motion (BWIM) for Different Types of Concrete Bridges <i>Jonas Geng</i>	Response of Tower Top Gantry under Construction Load <i>Changling Xie</i>
15:15		ML-Driven Strength Prediction and Mix Optimization for RAC <i>Chuanlei Zheng</i>			Structural Calculation and Adjustment Scheme Verification of Steel Cofferdam for Deep-Water Bridge Pile Cap <i>Changling Xie</i>

ROOM	306	307	308	309
13:45-15:30	[G06-7] Innovative structures and construction technologies: Recent projects in landmark and movable bridges structures	[G04-2] Sustainability, clean energy and carbon neutrality: Embodied carbon analysis & life-cycle assessment	[G05-7] Structural health monitoring, operation and maintenance: Smart sensors and field monitoring systems	[S06-1] Fragility and resilience estimation of infrastructure subject to natural disasters
Session Chair	Steve Kite	Rasmus Rempling	Sejin Kim	Do-Soo Moon, Young Joo Lee
13:45	The Unique Walyalup (Fremantle) Bridge <i>Lukasz Wojnarski</i>	Data-Driven Estimation of Bridge Life Cycle Maintenance CO2 Emission <i>Sanghyun Lee</i>	Robustness Assessment of Strain-Based Load Monitoring Procedures Applied to a Hybrid Rock-Concrete Underground Structure <i>Anna Teike</i>	Study on the Material Properties of Q355B After Exposure to High Temperature with Load <i>Xiaobin Qiu</i>
14:00	Tsing Lung Bridge, Hong Kong – Preliminary Design <i>Steve Kite</i>	Benchmarking and Reducing Upfront Carbon in Long-Span Bridges <i>Fumiya Murano</i>	Application of New Technologies in Structural Health Monitoring and Data Processing for Suzhou China Overseas Supertall Tower <i>Jiahui FAN</i>	Determination of Parameters for Dynamic Bridge Analysis: Specifics of Croatian Bridges and Condition-Based Adjustments <i>Ana Mandić Ivanković</i>
14:15	Special Isolation Bearings for India's First Vertical Lift-Bridge: The Pamban Bridge <i>Marcel Gruber</i>	Design and Research of Carbon Emission Calculation System for Cable-Supported Bridges <i>Jinfei Liu</i>	Structural Health Monitoring of Post-Tensioned Bridges Using Distributed Fiber Optic Sensors <i>Bartosz Piątek</i>	Assessing Bridge Seismic Reliability with Damage Indices: An FE-Coupled Computational Framework <i>Hyeonsung Song</i>
14:30	Danjiang Bridge – How Hydraulic Fuse Restraints Setting New Standards in Seismic Design <i>Gaetano Planese</i>	Climate Requirements Role in the Procurement of Infrastructure and Bridge Projects <i>Rasmus Rempling</i>	Sensor Strategies for Monitoring Stiffening Glazing <i>Nathalie Niesser</i>	Cyclic Behavior of Structural-Sealant-Glazing Systems under Seismic Loading <i>Tom Reisewitz</i>
14:45	Design and Construction of the New Grenland Bridge in Norway <i>Ziad HAJAR</i>		Field Monitoring and Control of Vortex-Induced Vibrations in Stay Cables of Cable-Stayed Bridges <i>Wei Gou</i>	Interaction of Stiffness and Mass Eccentricities in Reinforced Concrete Frame Systems <i>Do-Soo Moon</i>
15:00	Design and Construction of Godeok-Topyeong Bridge <i>Eun-Chul Lee</i>		Efficiency Enhancement of Energy Harvesters Utilizing Temperature Differences in Steel Deck Bridges <i>Kouchi Takeya</i>	Multi-Field Coupled Seismic Fragility Assessment of Power Plant Piping <i>Ziyu Wang</i>
15:15	Köhlbrand Replacement Crossing Hamburg – Project Setup and Inception <i>Henning Laubinger</i>		A Combined Denoising Method Based on GWO-VMD and Multi-Scale Principal Component Analysis <i>Can Liu</i>	Seismic Performance of Reinforced Concrete Walls with Minimum Vertical Reinforcement under Variable Axial Tensile-Compressive Loads <i>Keming Huang</i>



Physics-Informed Machine Learning for Crack Assessment in Edge-Cracked Beams Using PZT-based Scattering Coefficients

Nur Indah Mukharromah, Jin Ho Hwang, Hyun Woo Park

Department of ICT Integrated Ocean-front Smart City Engineering, Dong-A University, Busan, South Korea

Taejeong Lim

Korea Institute of Construction Technology, Goyang, South Korea

Contact: hwpark@dau.ac.kr

Abstract

Accurate crack assessment is essential for ensuring the structural integrity of engineering systems. Surface-bonded piezoelectric (PZT) transducers have shown strong potential for evaluating crack parameters. However, such data-driven crack assessment requires extensive experimental data that are costly and time-consuming. This study proposes a physics-informed machine learning (PIML) approach for crack assessment in edge-cracked beams that integrates PZT-based scattering coefficients using closed-form solutions. Closed-form PZT-based scattering solutions derived from Timoshenko beam theory, linear fracture mechanics, and one-dimensional piezoelectricity are used to generate a large training dataset through parameter sweeping of crack depth and excitation frequency in both pitch-catch and pulse-echo configurations. The trained PIML maps scattering responses to crack depth and validated through blind tests using finite element simulations.

Keywords: crack assessment; physic-informed machine learning (PIML); PZT-based scattering coefficients; edge-cracked beams; structural health monitoring.

1 Introduction

Structural health monitoring (SHM) has become a crucial instrument for ensuring the safety and integrity of engineering structures through damage identification [1]. Among various types of structural damage, cracks are particularly critical, as they can modify local stiffness and wave propagation characteristics, and may initiate catastrophic failures if undetected. Surface-bonded piezoelectric (PZT) transducers have been widely used in SHM to identify crack parameters such as depth and location due to their low cost and dual actuation-sensing capability [2]. PZT-based methods enable both pitch-catch (PC) and pulse-echo (PE) measurement configurations, allowing elastic wave interactions with cracks to be

captured through changes in transmitted and reflected signals [3].

Early studies primarily focused on detecting cracks through signal amplitude, time-of-flight, or frequency-domain features [4]. While effective for identifying the presence of crack, these indicators often lack a direct and quantitative relationship with crack parameters. Furthermore, such data-driven crack assessment approaches typically require extensive experimental data for calibration and validation. In practice, collecting and labelling large experimental datasets is often expensive, time-consuming, and sometimes infeasible. Moreover, the strong dependence on experimental data limits their scalability and robustness of these approaches, particularly under varying operational and environmental conditions.

More recently, wave scattering analysis has provided a physically interpretable framework for understanding crack-induced signal changes. Several studies have investigated wave scattering in cracked beams using beam theories such as Euler-Bernoulli [5], Timoshenko [6], and their combination [7]. Closed-form solutions for wave scattering coefficients have been derived to describe reflected and transmitted wave components as functions of crack depth and excitation frequency [8-10]. These closed-form solutions offer clear advantages in terms of physical interpretability and computational efficiency, and they provide a promising means to alleviate data scarcity and improve physical consistency. However, the explicit development and utilization of closed-form solutions for PZT-based scattering coefficients in crack assessment remain largely unexplored.

Machine learning (ML) techniques have been increasingly applied to crack assessment problems in SHM [11]. Despite their success, most existing ML-based approaches rely on large datasets obtained from experiments or finite element (FE) simulations. Physics-informed machine learning (PIML) provides a compelling alternative by embedding physical knowledge into the training process, thereby reducing data dependency. For wave-based SHM applications, incorporating physics-based data is particularly important for achieving accurate, data-efficient, and robust crack assessment. In the context of PZT-based crack assessment, closed-form PZT-based scattering solutions enable the generation of physically meaningful training data, making them well suited for PIML frameworks.

This study proposes a PIML approach for crack assessment in edge-cracked beams that integrates PZT-based scattering coefficients generated via physics-based data augmentation using closed-form solutions. The closed-form PZT-based scattering solutions is derived from Timoshenko beam theory, linear fracture mechanics, and one-dimensional piezoelectricity, and is valid below the beam cutoff frequency. By sweeping crack depth and excitation frequency under PC and PE configurations, a large physics-informed training dataset is constructed at minimal computational cost. The PIML approach is then trained to map

each input pair of excitation frequency and PZT-based scattering coefficients to crack depth. The proposed approach is validated through blind test on untrained datasets obtained from FE simulations for various crack scenarios.

2 PZT-based Scattering Model

Understanding wave scattering coefficient is a fundamental step in developing a PZT-based scattering model. In crack assessment using elastic guided waves, wave scattering coefficients serve as key parameters for characterizing the interaction between incident waves (A_1^+) and cracks. When incident waves, either axial (A_a^+) or propagating bending (A_p^+), encounters an edge crack, they are partially reflected (A^-) and transmitted (B^+) to three wave types: axial (a), propagating bending (p), and evanescent bending (e), as illustrated in Figure 1. Among A^- and B^+ , two waves arise from mode conversion, meaning that their wave types differ from that of the incident wave. The wave scattering coefficients are defined as the ratios of the amplitudes of the reflected (R) and transmitted (T) waves to that of the incident waves, and they quantitatively describe the crack-induced modification of wave propagation.

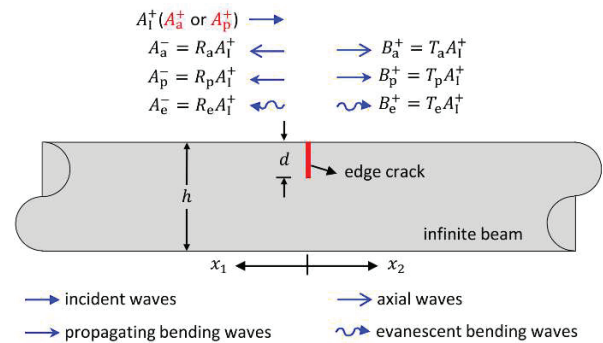


Figure 1. Wave scattering analysis

Park and Lim [8] developed closed-form wave scattering solutions under pure bending condition induced by A_p^+ . However, the accuracy of their formulation was limited to crack depth ratio (d/h) below 0.2. Mukharromah et al. [9] extended this work by incorporating axial-bending coupling (ABC) effects due to both A_a^+ and A_p^+ . This improvement extended the valid crack depth ratio range to approximately 0.3 for A_p^+ , although shear

behaviour was neglected. However, for A_a^+ , the numerical results were satisfactory compared to the FE simulations. Subsequently, Lim et al. [10] further refined the wave scattering formulation by accounting for axial-bending-shear coupling (ABSC) under A_p^+ . Their results showed close agreement with FE simulations, demonstrating that inclusion of shear behaviour significantly enhances prediction accuracy.

Similar to wave scattering analysis, in PZT-based crack assessment the interaction between the electrically excited signal and a crack modifies the received electrical response. The ratio between the received signal and the excited signal is defined as the PZT-based scattering coefficient, which provides a quantitative measure of crack-induced wave interaction. Figure 2 illustrates the PZT-based scattering analysis for both PC and PE configurations in an infinite beam. The host beam modelled using Timoshenko beam theory, which is essential for accurately capturing high-frequency wave propagation in SHM applications. The PZT transducers are modelled as wafer-type actuators and sensors that are perfectly bonded to the beam surface, with shear lag effects assumed to be negligible. The PZT dimensions, namely the length (2η) and thickness (h_{PZT}), are assumed to be small compared with the beam dimensions, allowing the effects of PZT mass inertia and stiffness to be neglected [12].

As depicted in Figure 2(a), the PC configuration use the left PZT as an actuator and the right PZT as a sensor to measure transmission coefficients (T_{PC}). In contrast, in PE configuration, as illustrated in Figure 2(b), the upper PZT acts as an actuator and the lower PZT serves as a sensor to capture reflection coefficients (R_{PE}). The actuator generates an incident wave in response to the applied voltage ($\bar{V}_{in}$) as the excited signal. The incident wave produces axial forces $\bar{F}$ and associated bending moments $\bar{M}$ at the edges of the PZT that propagates along the beam and interacts with the edge crack, where a portion of the wave energy is transmitted and reflected. These scattered wave components are subsequently captured by the sensor. Through electro-mechanical coupling modelled using one-dimensional linear piezoelectricity theory [13], the

sensor converts the resulting strain fields into measurable output voltages, denoted as $\bar{V}_{PC}$ for T_{PC} and $\bar{V}_{PE}$ for R_{PE} as the received signal.

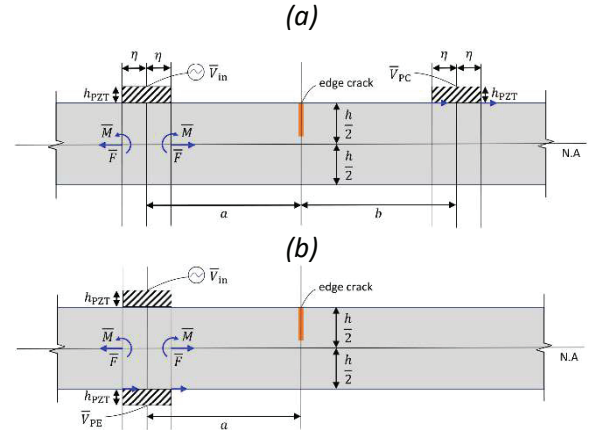


Figure 2. Edge-cracked infinite beam model with PZTs: (a) PC configuration; (b) PE configuration

Both $\bar{V}_{PC}$ and $\bar{V}_{PE}$ are proportional to the difference in axial displacements between the right and left edges of the host beam surface directly beneath or above the sensing PZT, depending whether the system operates in PC or PE mode, as illustrated in Figure 2. These axial displacements are obtained from the propagating transmitted and reflected wave components generated by the incident wave for PC and PE configuration, respectively. These displacement fields form the basis for defining the PZT-based scattering coefficients used as physics-informed features in the subsequent PIML framework. Assuming known distances between the actuating PZT and the edge crack (a), as well as between the edge crack and the sensing PZT (b), closed-form solution for the PZT-based scattering coefficients T_{PC} and R_{PE} can be derived. Due to the space limitation, the detailed derivation is not provided here.

3 Physics-Informed Machine Learning Framework

Machine learning and deep learning techniques have been increasingly adopted for damage assessment problems. However, such data-driven approaches typically require large amounts of training data, which are difficult to obtain in real-world mechanical applications. Moreover, purely

data-driven models often exhibit limited generalization outside the training domain. To address these limitations, PIML has emerged as a promising framework that integrates physical knowledge into the learning process [15]. Figure 3 illustrates the conceptual relationship between data availability and physics knowledge, highlighting how prior physics-based knowledge can compensate for limited training data by simulating the underlying physical behaviour of actual system.

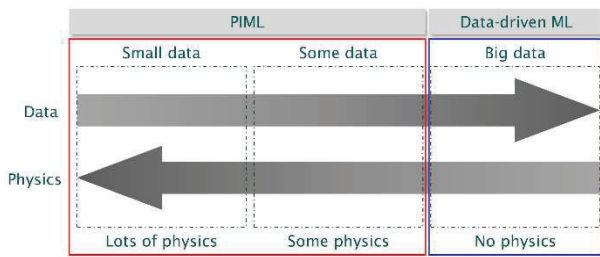


Figure 3. Relationship between data availability and physics knowledge [15]

Figure 4 presents the overall framework of the proposed PIML approach for crack assessment in edge-cracked beams. Closed-form PZT-based scattering solutions are used to generate physics-consistent training data. By sweeping crack depth and excitation frequency under both PC and PE configurations, a large augmented dataset is constructed at minimal computational cost. Each data sample consist of a pair of input features, namely the excitation frequency and corresponding PZT-based scattering coefficients, with crack depth ratio serving as the output label.

For randomly generated crack scenarios, all PZT-based scattering coefficient samples and their associated excitation frequency and crack depth ratio labels are generated through the PZT-based scattering model described in Section 2. The resulting dataset is subsequently provided to MATLAB's Classification Learner toolbox [16], where multiple machine learning algorithms are trained to map the input features to crack depth ratio. A model with the highest training accuracy is selected. Finally, the trained PIML model is statistically validated through blind test on previously unseen datasets obtained from FE simulations for various crack scenarios.

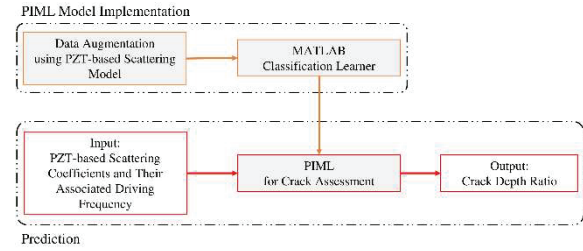


Figure 4. The proposed PIML framework

4 Validation

In this section, the implementation of the proposed PIML framework is described, followed by numerical validation results for crack depth ratio estimation. For the randomly generated crack scenarios, PZT-based scattering coefficient data are augmented for each crack depth ratio label. The input features for both classification and regression models consist of 53 discrete frequency-domain data points representing the mode-converted PZT-based scattering coefficients (T_{PC}^{ap} or R_{PC}^{ap}). Mode converted coefficients are specifically selected due to their high sensitivity to crack presence. These features are configured within a normalized frequency range of 0.2–0.6. This range is extracted from an initial broad frequency sweep ranging from zero to the host beam cutoff frequency (sampled with a step size of 2000 Hz to generate 134 raw features) to avoid regions where numerical instability or weak wave-crack interaction occur. For the classification task, the augmented data are categorized into three discrete crack depth ratio classes to distinguish between shallow, moderate, and deep crack conditions: Label 1 for crack depth ratios smaller than 0.2, Label 2 for crack depth ratios between 0.2 and 0.5, and Label 3 for crack depth ratios greater than or equal to 0.5. To enrich the physics-informed dataset, crack depth ratios were uniformly sampled from 0 to 0.7 with an increment of 0.005, resulting in 141 discrete data points. For each data point, the corresponding PZT-based scattering coefficients were computed using PZT-based scattering model.

The augmented data were provided to MATLAB's Classification Learner toolbox [16], enabling systematic training and comparison of multiple classification algorithms using the same dataset. All available classifiers in MATLAB are

evaluated under a five-fold cross-validation scheme [17]. Figure 5 presents the comparative evaluation of the tested models. To maintain the methodological simplicity and reproducibility, models requiring extensive manual tuning or additional configuration are excluded from consideration. The adopted evaluation criteria prioritize prediction accuracy as the primary factor for model selection. Specifically, models were first evaluated based on their average prediction accuracy and robust worst-case performance under varying training conditions. Among models exhibiting comparable prediction accuracy levels, computational cost was then considered as a secondary criterion. Based on the evaluation results, five representative models were selected for further validation using FE datasets: Medium Neural Network (MN), SVM Kernel (SK), Fine Tree (FT), Medium Tree (MT), and Coarse Tree (CT).

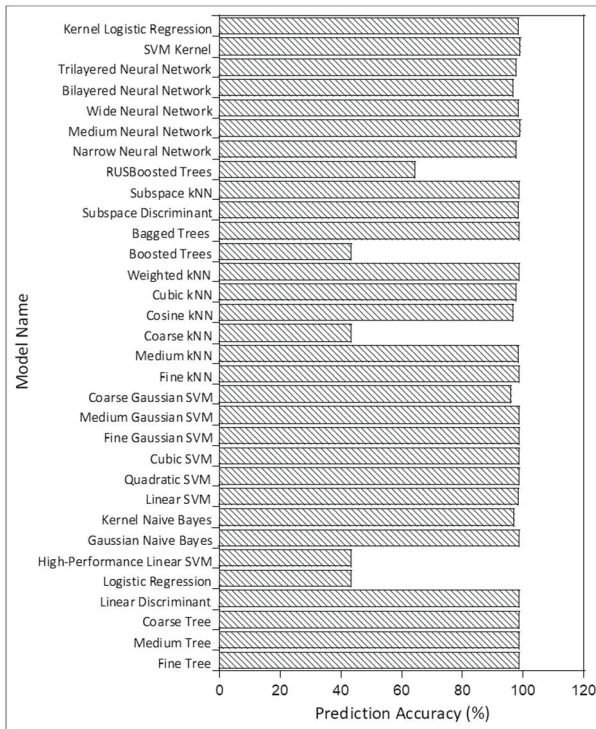


Figure 5. The comparative evaluation of the tested classification model

To validate the performance of the five selected models, blind test datasets were generated using FE simulation with surface-bonded PZT transducers, as illustrated in Figure 6. The base beam was modelled as aluminium with an elastic modulus of

70 GPa, a Poisson's ratio of 0.33, and a density of 2700 kg/m³. The PZT transducer had a thickness of 0.127 mm and material properties corresponding to PSI-5A4E. Table 1 summarizes the crack depth ratios used in the blind test datasets. Each test case corresponds to a specific crack depth ratio. Multiple tone burst (TB) excitation signals were applied for each case, except case 3, to further evaluate the reliability of the five selected models in predicting crack depth ratios from the FE blind test datasets. The excitation frequencies of the TB signals were selected within normalized frequency range used for training the PIML model to ensure consistency between the training and validation datasets.

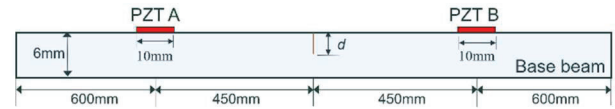


Figure 6. FE edge cracked beam model with PC configuration used to generate blind test datasets

Table 1. Validation results of the proposed PIML model for FE blind test datasets

Case	TB (kHz)	Crack depth ratio (Actual label)	Predicted Label				
			MN	SK	FT	MT	CT
1	100	0.167 (1)	2	1	1	1	1
	130		1	1	1	1	1
	160		1	1	1	1	1
2	100	0.333 (2)	2	2	2	2	2
	130		2	2	2	2	2
	160		2	2	2	2	2
3	160	0.5 (3)	2	2	2	2	2
4	100	0.667 (3)	2	3	3	3	3
	130		2	3	3	3	3
	160		2	3	3	3	3

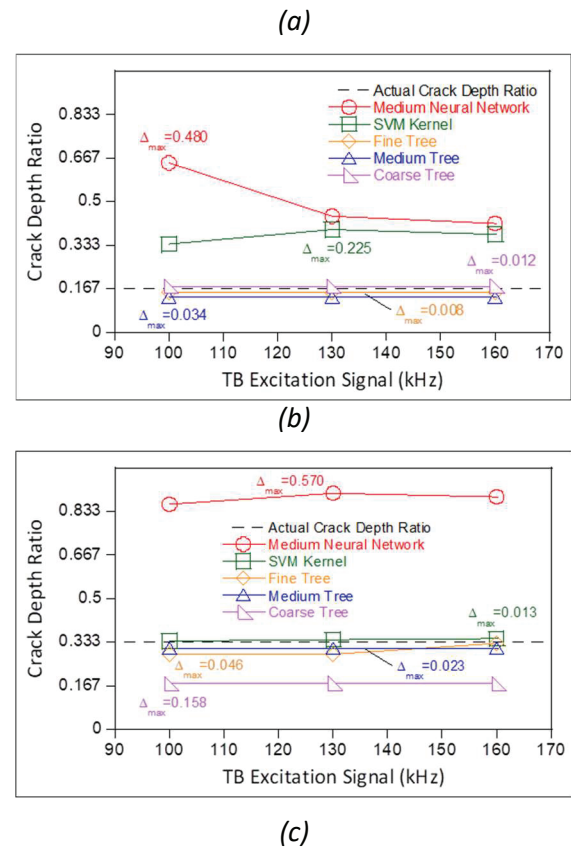
Table 1 also presents the validation results of the five selected models for each case under different TB excitation signals. The model predicts the crack depth ratio label for each FE blind test dataset according to the predefined classification scheme. For all the test cases, both the predicted labels and the corresponding actual labels are reported. As shown in Table 1, the SK and tree-

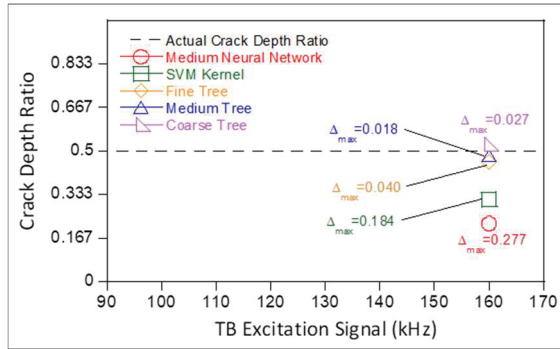
based classifiers (FT, MT, and CT) demonstrate consistent prediction performance on the FE blind test datasets, achieving 90% classification accuracy. In contrast, the MN exhibits less consistent behaviour in some cases, with an overall classification accuracy of 50%. These results indicate that the five selected models are generally capable of identifying crack depth ratio labels based on FE blind test datasets. The high classification accuracy observed for most cases, together with the absence of large misclassification errors, suggests that the extracted frequency-domain scattering features derived from physics-based model contain meaningful information related to crack severity. The misclassification observed in Case 3 does not necessarily indicate a limitation of the learning models, but rather reflects the inherent difficulty associated with samples located near the class boundary. Because the crack depth ratio of 0.5 lies exactly at the transition between Label 2 and Label 3, even small deviations in the predicted value may result in a different class assignment. This phenomenon is commonly referred to as boundary-class ambiguity.

However, classification results alone cannot fully indicate how accurately each model estimates the actual crack depth ratio. Therefore, an additional regression analysis was conducted to evaluate the models' capability to predict the crack depth ratio as a continuous variable, using MATLAB's Regression Learner toolbox [18]. In this analysis, the same physics-informed training dataset used in the Classification Learner was employed, and the five selected models were again selected for regression-based prediction.

Figure 7 presents the regression analysis results for each case in the FE blind test datasets. For Case 1, the FT, MT, and CT models produced the closest predictions, with estimated crack depth ratios of approximately 0.155, 0.133, and 0.175, respectively, all remaining within the correct Label 1 range. In contrast, the MN and SK models predicted values around 0.416–0.647 and 0.339–0.392, respectively, indicating significant overestimation. For Case 2, the SK, FT, and MT models achieved relatively small errors, predicting crack depth ratios of approximately 0.339–0.349, 0.288–0.330, and 0.310, respectively. However, the MN and CT models produced larger deviations,

with predictions around 0.529–0.570 and 0.175, respectively. For Case 3, which lies exactly on the classification boundary between Label 2 and Label 3, the FT, MT, and CT predicted values close to this boundary, around 0.460, 0.483, and 0.527, respectively. The slightly underestimated predictions from FT and MT explain why the Classification Learner assigned Label 2 instead of Label 3 for this case. Therefore, this misclassification arises from boundary proximity rather than a failure of the learning models. For Case 4, the FT, MT, and CT models again produced reasonable predictions within the correct Label 3 range, with estimated values of approximately 0.550–0.595, 0.573, and 0.527, respectively. In contrast, the MN and SK models significantly underestimated the crack depth ratio for both Cases 3 and 4. For Case 3, these models predicted values around 0.223 and 0.316, respectively, while for Case 4, the predicted values were approximately 0.276–0.364 and 0.340–0.369, respectively. This indicates sensitivity to domain shift between the training and FE blind test data.





(d)

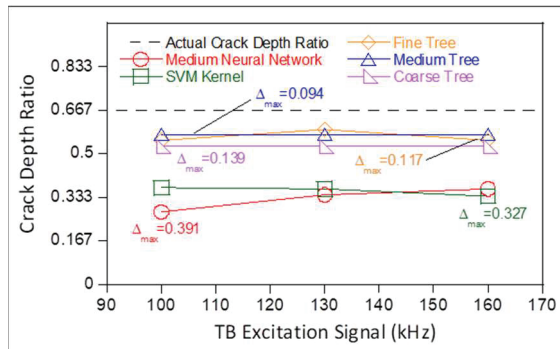


Figure 7. Regression analysis results of the proposed PIML model for FE blind test datasets: (a) Case 1; (b) Case 2; (c) Case 3; (d) Case 4

The continuous predictions demonstrate that the models are able to capture the underlying crack severity trend. The predicted crack depth ratios are consistent with the classification outcomes for approximately 40% of the cases for the MN and SK models, 100% for the FT and MT models, and 60% for the CT model. The predictions from the regression results may differ from the classification results because the Classification Learner assigns discrete crack depth ratio labels, while the Regression Learner estimates continuous crack depth ratios. When classification and regression outputs differ, classification is used as the primary estimate of the crack depth class, while regression checks if the predicted crack depth is close to an adjacent class boundary. In Case 3, the regression output confirms the prediction is close to the boundary, which is interpreted as boundary-class ambiguity rather than model failure. Overall, the regression analysis indicates that the tree-based models preserve the monotonic relationship between spectral response and crack severity, providing more physically consistent predictions

than the MN and SK models. Among the evaluated models, the MT model achieved the most reliable performance and was therefore selected as the most suitable model for crack depth ratio prediction.

5 Conclusions

This study proposed a PIML model for crack assessment in edge-cracked beams using PZT-based scattering coefficients. Closed-form scattering solutions derived from Timoshenko beam theory, linear fracture mechanics, and one-dimensional piezoelectricity were used to generate physics-consistent datasets. By sweeping crack depth ratios and excitation frequencies, a physics-informed dataset capturing the frequency-dependent scattering characteristics was constructed. Among the evaluated algorithms, the MT model provided the best performance. Validation using FE blind test datasets with surface-bonded PZT transducers under different TB excitation signals showed 90% classification accuracy in identifying the crack depth ratio labels. Additional regression analysis further demonstrated that the model can estimate continuous crack depth ratios with relatively small errors, confirming the effectiveness of the physics-informed scattering features for reliable crack assessment.

Overall, the results demonstrate that the proposed PIML model provides a promising and computationally efficient approach for crack depth evaluation in wave-based SHM systems. By leveraging physics-based scattering models to augment training data, the framework reduces the reliance on large experimental datasets while maintaining robust predictive performance. Although the proposed model was validated using FE blind test datasets, a numerical-experimental gap is expected. Factors like environmental and sensor noise, varying PZT bonding conditions, and modeling uncertainties can influence the measured datasets and reduce estimation accuracy. Future work will include experimental datasets and noise-contaminated training cases to mitigate this gap.

6 References

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